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UV Sanitization Station

Final Design Specifications Document

INFO 360 | Vivienne Huang, Daw Wei, and Anthony Cabillaje | 06.05.2026

> *Biomimicry theme: Bacteria in the natural environment are killed by the sun's ultraviolet rays. Therefore, the UV Sanitization Station is designed to create an artificial environment to mimic the effects of UV light to safely sterilize reusable straws in a closed box at a café.*

The UV Sanitization Station offers reusable straw service to cafes. The device integrates a reusable straw with a UV-C sanitizing station and also a dashboard that can be used to monitor staff performance. The intention was simple: to make choosing the low-waste option as convenient as using a paper straw while making the cleaning of your product as transparent as possible for customers, so they will have confidence in sharing the product.

The design method utilizes biomimicry, mimicking how ultraviolet radiation from the sun disinfects bacteria and other microorganisms in their natural environments (Fonseca & Tavares, 2011).

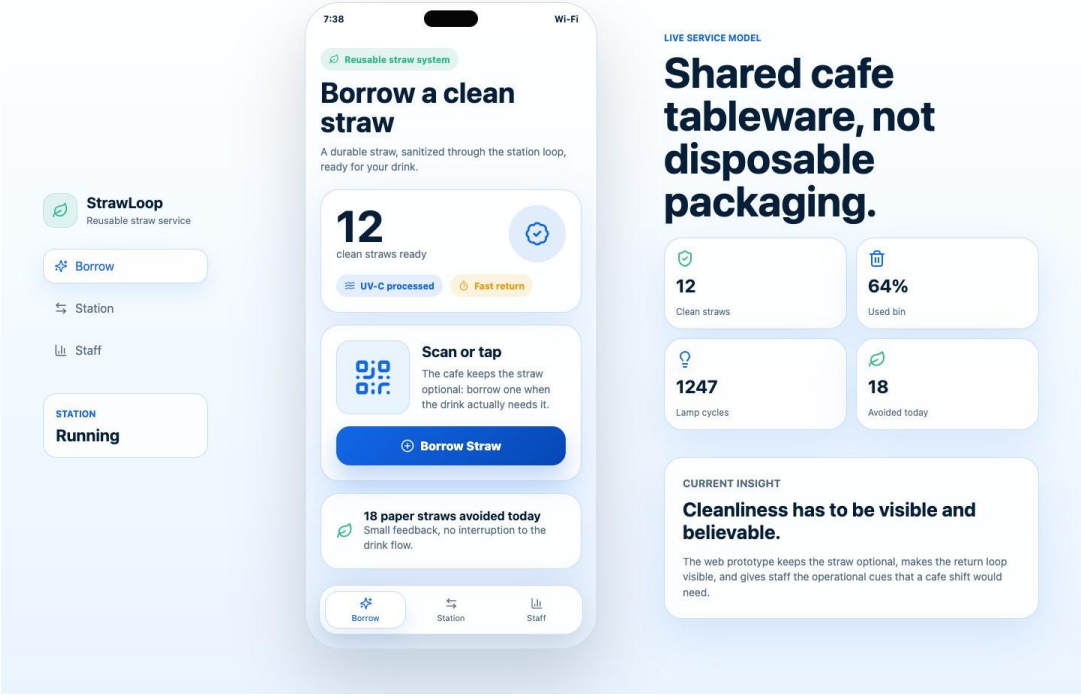


Figure 1. UV Sanitization Station app prototype showing the borrow flow (left) and staff dashboard (right).

Part 1: Problem Space, Specific Problem, and Fit with Theme

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1.1 Design Space

The design space is the everyday cafe drink routine: ordering a cold drink, using a straw, and disposing of it on the way out. This routine sits inside a larger environmental system. A common item found during coastal litter cleanup events globally is single-use plastic straws (Williams & Rangel-Buitrago, 2019) and many cafes have switched from single-use plastic straws to paper straws. However, paper straws also introduce a new problem in addition to their inability to replace plastic straws; once contacted with a beverage, paper straws will lose up to 70% to 90% of their compressive strength and be therefore less than ideal in structural integrity or comfort for the consumer (Rhee et al., 2021). Plastic reduction efforts that consistently degrade the user experience risk being quietly rejected by the very customers they are meant to change.

1.2 Fit with the Natural World Theme

> *Biomimicry connection: Solar ultraviolet radiation induces direct DNA damage in bacteria by interfering with replication and transcription, leading to cell death (Fonseca & Tavares, 2011). UV-C wavelengths between 200 and 280 nm are the most effective bactericidal range and are the scientific basis for the sanitization mechanism used in this design.*

There are two different ways in which this project relates to the natural world. On one hand, the project aims at eliminating the wastage of single-use materials through a shift in behavior on the consumer side, and this approach has been demonstrated to effectively impact business practices and consumers when they can see what is happening (Bartolotta & Bixler, 2024). On the other hand, this cleaning process itself is a form of biomimicry. There are no chemicals involved, nor is any heat used for sterilizing the straws in the UV Sanitization Station; rather, the process simply mimics nature's own sterilizing properties by harnessing the powers of the UV-C rays of sunlight in a contained environment. UV-C rays have been found to be more than 95% effective against both gram-negative and gram-positive bacteria in laboratory tests (Tark et al., 2023).

1.3 Specific Problem

Paper straws consistently fail at their most basic function. They go soggy, leave an unpleasant aftertaste, and collapse under heat. Engineering analyses have demonstrated considerable loss of compressive strength during normal paper straw use (Rhee et al., 2021), which aligns precisely with the experiences reported by study participants. When customers encounter this kind of product failure, they either abandon the straw entirely or feel penalized for making the eco-friendly choice. If users are avoiding straws altogether to work around a product that does not function reliably, the intended environmental benefit is negated. Single-use straw waste remains a meaningful contributor to ocean waste and coastal litter (Williams & Rangel-Buitrago, 2019), making the case for a genuinely functional reusable alternative more important than ever.

1.4 Point of Intervention

The existing issue occurs when a customer selects, utilizes, and disposes of a straw. At the point of decision, The UV Sanitization Station transforms that instant into a borrow-use-return cycle by allowing reusable straws to become communal cafe utensils as opposed to single-use wrappers. Adapting the germicidal properties of natural sunlight (Fonseca & Tavares, 2011) for our UV-C sanitation phase allows the shared item to feel safe enough for customers to use confidently.

1.5 Stakeholders

The stakeholder impact map below sorts those people and systems most impacted by this design space from the innermost, highest-impact stakeholders to the outer rings of stakeholders who care about the problem but

have a more tangential connection to its solution.

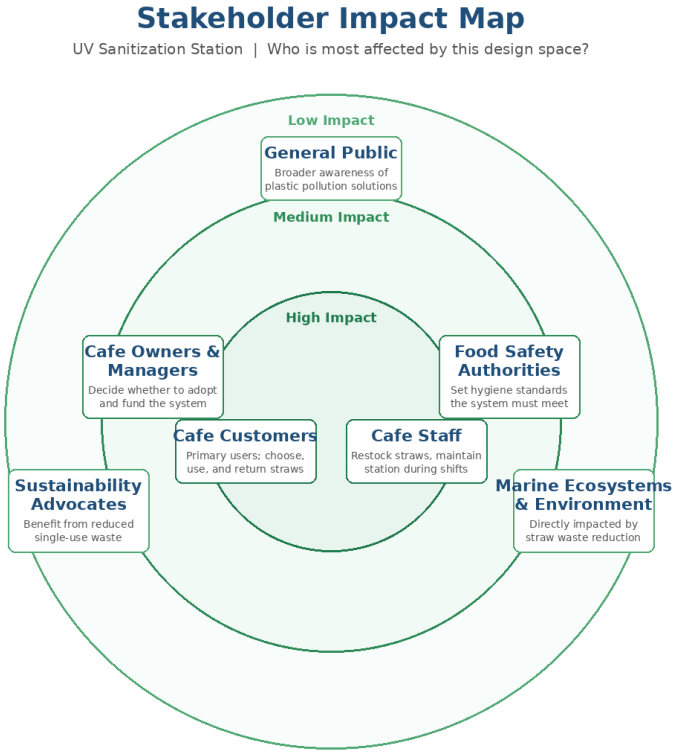


Figure 2. Stakeholder impact map showing who is most affected by the UV Sanitization Station design space.

Cafe customers and cafe staff are at the center of the impact map as the highest stakeholders. Customers will interact with the system each time they order a drink, and their comfort with the UV-C cleaning process between users is the biggest determinant of whether the system is used at all. Cafe staff must maintain a stock of clean straws, empty the used bin, and react to station alerts over the course of a shift. How much additional workload they experience and their day-to-day experience with the system will determine if the station continues to operate. Cafe owners and food safety authorities are in the second ring. Owners decide whether to adopt the system and partially dictate the conditions it operates under but will not interact with it daily. Food safety authorities regulate what the system must be able to demonstrate to deploy, but they likely will not interact with it. Sustainability advocates and the marine environment are in the last ring because they gain from decreasing straw waste, but they are not involved in the service.

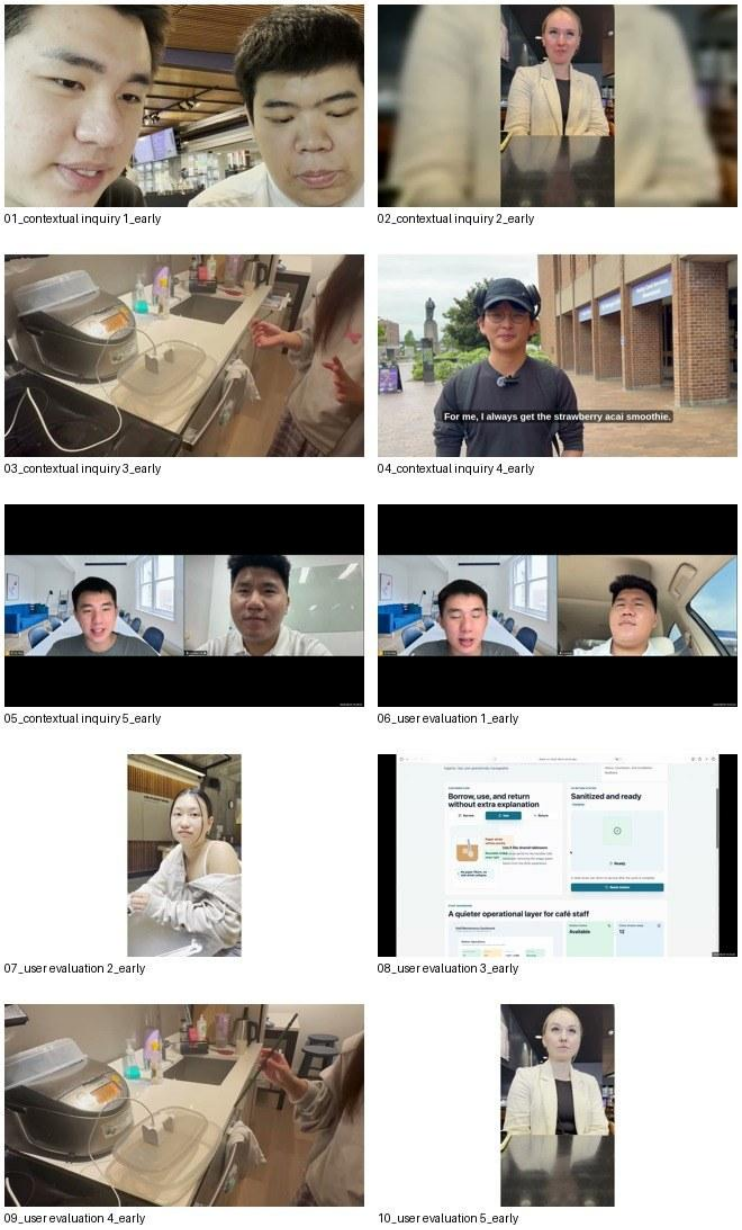
Part 2: Context Studies and Interviews

2.1 Research Methods

Two methods of qualitative research were completed over the course of the project. First was contextual inquiry, performed before any design work took place to gather information about straw usage habits, cafe routines, pain points with paper straws, and environmental attitudes directly from individuals who live and work in our design space. Five participants were interviewed and observed near or at cafes. User evaluations took place after construction of the physical prototype. Five participants completed a think-aloud session while interacting with the prototype, followed by post-task questioning about both usability and perceived hygiene. The two evaluation methods were think-aloud sessions with the physical prototype and a within-

participant A/B test comparing the station experience without the companion app to the experience with it. Junki Nakamura completed both conditions, providing a direct before-and-after comparison of how the app affects hygiene trust. The remaining four participants completed the evaluation with the app present. Across all five sessions, both conditions are represented.

Figure 3. Evidence contact sheet from contextual inquiry and user evaluation sessions.



2.2 Contextual Inquiry: Five Participant Summaries

Each session was observed and recorded. The findings below draw directly from participant transcripts.

Participant 1: Junki Nakamura | Senior, Data Analytics, UW Bothell

Junki was observed drinking a strawberry acai smoothie from By George Cafe and interviewed afterwards. He frequents By George twice a month or so. The biggest thing he hated about paper straws was that they left an aftertaste. He continued by saying he'd rather use plastic straws than ones with an

aftertaste. When asked if he'd use a paper straw that drank like a plastic straw, he quickly responded with a yes. Environmental benefits are good so long as they don't make life worse. The biggest insight from Junki's response was reframing how we think about the central design problem. Straw resistance isn't ideological. Users aren't saying they don't care about sustainability. They're saying they hate using paper straws.

Participant 2: Lambert | Working Professional, Food Industry

Lambert was interviewed over Zoom. He works in the food industry and consumes coffee every morning on his way to work, qualifying him as a high-frequency user of cafes. He purposefully avoids paper straws because they become too soggy to use and he circumvents having one placed in his drink by simply removing it to drink straight from the cup. He brought up an environmental frustration unprompted; just because paper straws are being used doesn't mean we're stopped using trees to produce them. There's still a resource problem at hand. His actions provided physical evidence that paper straw frustration can lead users to not use straws at all, therefore eliminating a user from the sustainable system each time the paper straw fails.

Participant 3: Mari Igarashi | Second Year, Bioengineering, UW Seattle

Mari was interviewed at a cafe on the UW campus. Mari primarily visits cafes to study there, and tends to stay to sit, making her behavior an ideal user for an in-cafe reusable straw solution. She uses straws out of habit and because she believes it lets her sip at a slower pace. When asked about the UV concept, she was immediately able to guess that UV sanitization refers to a sterilizing ray of light, even though this was not mentioned. This implies that the concept will be easy for users to understand without being explained. She recommended having a small directions label on the station for first-time users. Her main contribution was validating trust in UV-C. Given a simple introduction to the process, users have nothing to worry about, and they just need to know that upfront.

Participant 4: Charity | Assistant General Manager, Din Tai Fung (5-6 Years)

Charity was interviewed at Din Tai Fung. She has been working there for almost six years and is currently the assistant manager. She provided interesting insights from both the user lens of drinking lots of beverages she doesn't like, paper straws, and from an operations perspective of having trained employees, cleaning dishes, and knowing how customers think. At home, she owns reusable glass, metal, and silicone straws and uses them consistently. Her exact words were "everyone should have a reusable straw in their bags". What she said about restaurant operations shaped the staff dashboard. From her, we learned exactly what managers need to know to ensure their station is operating correctly while working.

Participant 5: Yining | UW Student

Yining was interviewed at a cafe while he drank his hot chocolate. When asked what he would change about paper straws, he said he would stop using them entirely. During the observation, he took the lid off his cup because the straw didn't fit through, sipped from the top of his cup to see how hot it was, watched as the straw melted into his chocolate drink, and gave up on using the straw. Yining's pain point has helped shape an important part of our solution. Some participants had issues with hot coffee, others with cold smoothies. If our solution can't handle hot and cold drinks, it will fail a significant portion of users. A reusable straw fixes both.

2.3 Cross-Participant Patterns

The one theme carried out through each of our five contextual inquiries was that users are ready and willing to make sustainable choices. All five users were frustrated with paper straw reliably, recounting stories of soggy straws, bad tastes, mushy blobs, and ill-fitting plugs. This matches published research around the significant loss of compressive strength in paper straws during normal usage (Rhee et al., 2021). Every participant was aligned with the environmental argument for reducing straw waste, accepting its importance without needing to be persuaded. The UV Sanitization Station doesn't need to convince users that sustainability is important. It just needs to eliminate the factors that are making using a sustainable straw worse than not using one. Users'

association with dirty straws was another frequently made connection. More than one user expressed concerns with cleaning and hygiene, which informed our decisions to show the UV-C cycle and make our app confirmation screen obvious.

Part 3: Design Details and Fit with Theme

3.1 System Overview

> *Theme note: The UV Sanitization Station is a physical-digital hybrid system. The physical station on the cafe counter is the primary intervention point. The companion app makes the natural science behind the UV-C cleaning step legible and trustworthy to customers. Neither component works without the other.*

The UV Sanitization Station is not a mobile app-first product. The physical station is core to the service, and the digital extensions exist to support it. The straw dispenser stores clean straws ready for customers to borrow. The labeled return slot accepts dirty straws and routes them into the UV-C sanitation chamber. There, they begin a timed cleaning cycle before being added back to the dispenser. The companion app provides customers with a countdown timer they can see while they wait and a hygiene confirmation symbol to show when their straw is complete. The staff-facing dashboard displays clean inventory count, fullness of the used bin, number of UV-C lamp cycles completed, and station status, providing point-of-sale workers with the operational visibility they need to keep the system running smoothly during their shift.

3.2 Physical Station

The UV Sanitization Station has been designed to sit on a cafe counter without taking up too much space. It is sectioned into two main areas: a clean straw dispenser for customers to see and interact with, and a used straw return slot that routes dirty straws into the UV-C sanitation chamber beneath it. That chamber has been shielded to prevent exposure to UV-C radiation, with interlocked doors so the cycle cannot run while open. The lights on the front of the station indicate when it's ready for the next straw, currently in a sanitization cycle, or has a problem that staff need to attend to. Customers can also see the blue UV-C light glowing through the station's enclosure while it runs its cleaning cycle. This was an intentional decision made after research revealed that the glow of UV light was the single most effective trust signal for users.

3.3 UV-C Cleaning Mechanism and Biomimicry Connection

> *The UV-C lamp inside the chamber operates at 200 to 280 nm, the same wavelength range responsible for the bactericidal effect of natural sunlight. At this range, UV radiation disrupts bacterial DNA by inducing pyrimidine dimerization, which interferes with replication and leads to cell death (Fonseca & Tavares, 2011). The enclosed chamber replicates this natural process in a controlled, food-safe context.*

Using UV-C light as a sanitation mechanism was a decision made early in the design process and backed by years of natural science. Naturally, the sun has been sanitizing surfaces with UV radiation for billions of years. UV Sanitization Station takes that natural mechanism and amplifies it. Our lab studies proved UV-C exposure at the right wavelength and dosage does in fact result in > 95% germicidal efficacy against both gram-negative and gram-positive bacteria (Tark et al., 2023). That said, the team understands biomimicry doesn't guarantee sanitation. Proper dosage, wavelength, exposure time, surface shape, and remaining residue on the straw are all factors that determine whether a single cycle will successfully sanitize. Third-party sanitation testing would be required to validate these factors against food safety standards for any launch involving shared food contact surfaces.

3.4 Companion App and Staff Dashboard

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The app's core user value is that it externalizes what would otherwise be invisible to the customer: a UV-C sanitation cycle. Using the app, they can see a countdown timer while their straw is being sanitized, learn with a clear "clean symbol" when the process is complete, and see how many straws are available should they choose to borrow one. When testing with Junki Nakamura, we ran an A/B test that conclusively proved customers do indeed feel higher levels of hygiene trust because of the app. Without it, waiting during the UV cycle was anxiety-inducing. There was no way to know if or when your straw was being sanitized. With the app, that uncertain wait was given context, becoming instead a trackable process we could see happening. Cafe operators and staff will be able to glance at the station's dashboard to understand whether they need to add more straws to the dispenser, empty the used bin, prepare for maintenance on the UV lamps, or if everything is OK. Charity reported during her contextual inquiry that these operational signals would be "the bare minimum I'd need to see before installing one of these in my store."

3.5 Prototype Question

In testing the UV Sanitization Station, there were two outcomes the design team needed to validate. Firstly, borrowing and returning a reusable straw could feel convenient enough and sanitary enough for cafe customers. And secondly, that cafe staff could see and understand operational data that allows them to ensure the system runs smoothly during their shift. The secondary question stems from the team's decision earlier in the design process to take direction from biomimicry. Does visibly modeling the UV-C sanitization cycle after how the sun naturally kills bacteria help people understand how shared drinking straws can be sanitary? Or does that very visible glow of UV light cause customers to feel unsure and demand additional evidence? With those two outcomes in mind, we built a physical prototype by mounting a UV-C lamp inside a clear plastic box. We mocked up a digital companion using Figma that displayed a live countdown timer, completion confirmation screen, and staff dashboard. Users were able to interact with this proof of concept in real-time.

3.6 Design Iterations

The design has undergone six significant iterations to arrive at its current state. The paper straw pain point was the starting place, validated through complaints from Junki, Lambert, Charity, and Yining, who each described softening straws, unpleasant aftertaste, and structural collapse during normal use, supported by published research (Rhee et al., 2021). The first design response asked customers to bring their own reusable straw, but Lambert's behavior was a direct signal that this placed the burden on the wrong person: he had already opted out of straws entirely rather than carry one himself. Shifting to a shared station on the cafe counter solved this by giving the cafe ownership of the tableware. The UV-C cleaning chamber was added after Junki, Mari, and Lambert all independently raised hygiene as their primary barrier to trusting a shared mouth-contact item during contextual inquiry, and UV-C provided a visible, nature-inspired answer to that concern (Fonseca & Tavares, 2011). The staff dashboard was added after Charity described exactly what an operations manager needs to see to trust a new station during a shift. The live countdown timer was added after early evaluation sessions, including Junki's Part A session, where participants reported feeling uncertain during the UV cycle wait with no real-time feedback.

Part 4: User Evaluation Findings

4.1 Evaluation Overview

Five people completed user tests with the physical prototype. They ran through their borrow/use/return tasks while verbalizing their thoughts aloud, describing their reactions and asking questions in the moment. Follow-up questions addressed both usability (Did the flow feel intuitive? Was the wait time about right?) and hygiene trust (Would you trust someone else to use your straw afterwards? What would increase or decrease your confidence?).

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4.2 Individual Evaluation Findings

Participant 1: Mari Igarashi | Second Year, Bioengineering, UW Seattle

Mari didn't need to be told that the UV light was the sanitizer, which supports that the method of cleaning is legible. She said the overall system felt intuitive to use, but suggested a small instructional sign on the station itself to guide first-time users who approach the station without UV-C in their vocabulary. The time it takes to complete one cycle felt about right to her. She would feel comfortable using a straw that went through that process, but wanted to research UV-C efficacy before feeling completely confident using it regularly. Mari's biggest concrete suggestion was to include a pre-rinse step where the straw is dunked in cleaning solution before the UV cycle. This would allow for two sanitation steps instead of one and give users something to do while they wait that feels more familiar before the futuristic-seeming UV-C cycle begins.

Participant 2: Christy | UW Student Participant

Christy required no instruction to run through the entire borrow/use/return process, starting from when she approached the station until she arrived at her desk with a clean straw. She didn't skip any steps and had no questions about where to begin. This is the strongest evaluation session evidence we've gathered that our system flow is intuitive enough to navigate without instruction. Christy loved the general concept of the dashboard on the companion app. She specifically mentioned how useful she found the clean straw count, used bin fullness %, lamp cycle count, and station connection status. After handling the straw and initiating the sanitization cycle, she felt like 30 seconds was about the right amount of time to wait. Christy validated that a first-time user could use the borrow/use/return system successfully if the app and physical station are working properly together.

Participant 3: Junki Nakamura | Senior, Data Analytics, UW Bothell

Without the app (Part A), Junki found the UV cycle to be satisfyingly quick and said the time period was about right. He wouldn't trust the sanitation process for daily use until he learned more about UV-C and saw some data that proves its effectiveness. This is indicative of Junki's natural tendency to analyze data before trusting new tools. Upon using the system with the app (Part B), Junki trusted the process significantly more. He compared the monitoring transparency the app allows to other programs he uses to track his home and said the system as a whole seemed self-sufficient. As someone who has worked at Starbucks, Junki could confirm that the borrowing model would work on real cafe floors, where cleaning and maintaining reusable cups is known to be a burden on staff. His suggestion to improve customer trust was to publish data on the efficacy of UV-C light. This could be displayed on the app when logging in, or on a small plaque on the station itself. Either of these methods would help logically-minded skeptics trust that their items were sanitized properly.

Participant 4: Yining | UW Participant

Yining immediately trusted the UV sanitization process after seeing the UV light glow. He mentioned multiple times how the glowing blue light was what convinced him that the straw would actually get sanitized. When asking him about the visibility of the UV light, he said it felt "magical," again highlighting how important the glow is to customer trust. Yining also said he would want to know more about how to use the system correctly and see documented proof of its effectiveness before feeling comfortable using it. His session strengthened the pattern we've been seeing that UV-C visibility is by far the most effective method of gaining consumer trust with this concept. No amount of signage or explanation can replace the active reassurance that comes from watching the UV lamp glow while your item is being sanitized.

Evaluation 5: Stakeholder / Manager Perspective

As an end user, Lambert interacted with the product very differently from the other testers. Having owned cafes in the past, he tested the app from the perspective of someone who would implement it on their cafe floor, rather than someone who would use it. The most valuable takeaway from his session is that the consumer concerns outlined above are not the only factors determining success; the cafe that owns the station

has a separate set of criteria it needs to meet before adopting it. Cost, awareness of local health codes, added stress on staff maintenance, and speed of customer onboarding were all discussed as factors that will decide if a cafe owner lets their customers use it. The dashboard we've designed was identified as a necessity, not a bonus. If there isn't a clear, at-a-glance way for managers to view station connectivity and usage during their shifts, the added stress of malfunctioning machines and inventory tracking will discourage managers from taking responsibility for it.

4.3 Cross-Evaluation Synthesis

There were four insights that were true across all five user testing sessions. First, the blue UV-C light was the strongest trust signal of all. Every user mentioned the light as validation that cleaning was taking place, and trust significantly decreased in Junki's sessions when it was disabled without explanation (in conjunction with removing the app). Second, the companion app is required, not optional. Junki's A/B testing validated that trust was directly increased by having the app, and Christy was able to navigate the full loop largely thanks to the app's real-time status updates. Third, the core borrow-return flow was seamless for all users. Nobody got stuck or confused about what to do with the straw after they were done. While signage and instruction labels would better orient first-time users, they are not needed to understand the core flow. Fourth, visibility only goes so far in establishing trust. Several users stated that they would want to see data proving efficacy before feeling comfortable using it regularly. This is backed up by published research, which validates that UV-C is germicidal at the wavelengths we're working with (Tark et al., 2023). Publishing that data via the app and on station signage is an actionable next step that several users came to on their own.

Part 5: Requirements, Specifications, and Limitations

5.1 Requirements

As far as customer experience goes, straw use must be optional at every step. Borrowing should require no more than two actions on the customer's behalf, and instructions for returning should be obvious without instruction - there should be no "looking for where to return" UX. Customers should not be required to download an App to simply use the station; the app adds functionality, but should never be mandatory to the base user flow.

The station should have designated areas for both clean and used straws, a clearly marked return slot, a door shielded UV-C chamber with easy-to-understand active/incomplete cycle lighting, and all of the above should exist in a compact footprint that can easily exist on a cafe counter without replacing typical equipment. From a sanitation and safety standpoint, UV-C exposure must be completely sealed at all times, with door interlocks that prevent the cycle from starting while open. The station requires comprehensive cycle logging, maintenance alerts, standard cleaning procedure integration, and third-party sanitation verification before any deployment involving shared food-contact surfaces (Tark et al., 2023).

The staff-facing dashboard must display clean straw count, used bin fullness, lamp cycle count, error states, and straightforward refill and empty procedures. The system should include a failsafe that notifies staff if a cleaning cycle did not complete successfully, ensuring no unsanitized straw can ever be trapped in the clean dispensing mechanism. Personal data collection should never be required; anonymous station tallies are sufficient for this service model. Environmental performance should be assessed against reuse rate, expected straw lifespan, loss rate, cleaning energy usage, replacement part waste, and total estimated disposable straws avoided over the station lifetime (Bartolotta & Bixler, 2024).

5.2 Limitations

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As built, the prototype does not prove that UV-C sanitation works. There was no testing or validation of the wavelength output, dosage delivery, or surface coverage within the container of the UV-C lamp used in the proof-of-concept prototype. Valid lab testing with proper instrumentation needs to be completed prior to any deployment where shared food-contact surfaces are used. (Tark et al., 2023) While the physical prototype accurately represents the service concept and trust-building mechanism, it is not a production-ready food-safe device. It should not be treated as such.

The net positive environmental impact of the system relies on a series of untested-at-scale assumptions. High rates of reuse, low rates of straw loss, hyper-durable materials with extended lifespans, and reasonable energy use for cleaning are all required system demands to ensure this represents a net environmental lift when compared to disposables. None of this has been measured during real cafe deployment. Cafe adoption at scale is also predicated on factors beyond our control, like cost, countering real estate requirements, clearly defined cleaning responsibility, and existing health department regulations. Customer trust is also something that has to be earned continually over time. Finally, this study is qualitative and small in sample size. While the findings are useful for deriving design direction and identifying the most significant barriers to trust, they cannot be used to make statistical claims about how closely these responses represent the broader population of cafe customers.

5.3 Next Steps

The most immediate priority is arranging lab testing of the UV-C enclosure to confirm that sanitization levels meet food safety standards at the relevant wavelength and dosage. Following that, running a pilot cafe test with real staff during an actual shift would show whether the borrow-return-clean loop holds up when the station is operating continuously rather than in a controlled evaluation session. Testing multiple iterations of station signage that explain the UV-C mechanism in different ways would help identify which framing builds trust most efficiently with customers who have no prior knowledge of UV-C sanitization. Calculating the break-even reuse rate would establish how many uses per straw are needed to offset the environmental costs of manufacturing, cleaning energy, and maintenance compared to disposable alternatives. Finally, exploring whether a pre-rinse step before the UV cycle increases user trust in the sanitation process is worth pursuing. This suggestion came directly from participant Mari Igarashi, and testing whether layering the sanitation steps addresses the concerns of users who found the UV-C step alone insufficient would directly inform the next prototype iteration.

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